25901207

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Skill Enhancement Course (SEC) - MG4SECCFS200 - PROGRAMMING INJAVA
(2024 ADMISSION ONWARDS)

Part A – Short Answer Questions (1 Mark Each)

1. Differentiate between Class and Object. [CO1]
- **Class:** Blueprint/template for creating objects.
- **Object:** Instance of a class containing actual data and behavior.

2. Distinguish between Post-decrement and Pre-decrement operators. [CO1]
- **Post-decrement (x--):** Uses current value, then decrements.
- **Pre-decrement (--x):** Decrements first, then uses the value.

3. Why is Java a Robust Language? [CO1]
Java is robust because it provides:
- Strong memory management
- Exception handling
- Automatic garbage collection
- Type checking

4. When does the finally block execute in Java? [CO2]
The finally block executes after the try-catch block regardless of whether an exception occurs or not (except in abnormal JVM termination).

5. Examine the use of the start() method in threads. [CO2]
The start() method creates a new thread and invokes its run() method, enabling concurrent execution.

6. Differentiate between JCheckBox and JRadioButton. [CO3]
- **JCheckBox:** Allows multiple selections.
- **JRadioButton:** Allows only one selection within a ButtonGroup.

7. Examine the purpose of KeyListener. [CO3]
KeyListener is used to handle keyboard events such as key pressed, released, and typed.

8. Differentiate between init() and start() methods. [CO3]
- **init():** Called once when an applet is loaded.
- **start():** Called after init() and whenever the applet becomes active.

9. Why is main() method declared static? [CO1]
Because the JVM invokes main() without creating an object of the class.

10. Define Multilevel Inheritance with syntax. [CO1]
Multilevel inheritance occurs when a class inherits from another class, which itself inherits from another class.

class A {}
class B extends A {}
class C extends B {}

11. Write the size of a float data type. [CO1]

4 bytes (32 bits)

12. Use of setBounds() method for a Swing component. [CO3]

setBounds() specifies the position and size of a component.
button.setBounds(50,50,100,30);

Part B – Short Essay Questions (5 Marks Each)

13. Describe the concept of labelled loops. [CO1]

A labelled loop is identified with a label name and is mainly used with break and continue statements.
outer:

Advantages

- Improves control over nested loops.
- Allows exiting multiple loops at once.

14. Explain the exception handling mechanism in Java and flow of control. [CO2]

Exception Handling Blocks

- try
- catch
- finally
- throw
- throws

Flow of Control

1. Statements in try execute.
2. If exception occurs, control transfers to matching catch.
3. finally executes.
4. Program continues after exception handling.

Eg:

15. Demonstrate the use of JTextArea in Swing. [CO3]

JTextArea allows users to enter multiple lines of text.
Eg:

Features

- Multi-line text input.
- Supports text editing.

16. Define Visibility Control. List different visibility controls in Java. [CO1]

Visibility control determines accessibility of class members.

Access Modifier	Same Class	Package	Subclass	Other Package
Private	Yes	No	No	No
Default	Yes	Yes	No	No
Protected	Yes	Yes	Yes	Yes

Public	Yes	Yes	Yes	Yes

17. Contrast drawRect() and fillRect() methods in Graphics. [CO3]

drawRect()	fillRect()
Draws rectangle outline	Draws filled rectangle
Only border visible	Entire area colored
Less memory use	More graphical effect

```
g.drawRect(10,10,100,50);
g.fillRect(10,70,100,50);
```

18. Explain the role of default case in switch statement. [CO1]

The default case executes when none of the cases match the expression value.

```
int day=8;

switch(day){
case 1:
  System.out.println("Monday");
  break;
default:
  System.out.println("Invalid Day");
}
```

Importance

- Handles unexpected inputs.
- Improves program reliability.

Part C – Essay Questions (10 Marks Each)

19. Explain Constructor Overloading with Example. [CO1]

Constructor Overloading

- Multiple constructors with different parameter lists in the same class.
- Enables object initialization in different ways.

Eg:

Advantages

- Flexibility in object creation.
- Improves code readability.
- Supports multiple initialization methods.

20. Different Ways of Creating Threads in Java and Their Suitability. [CO2]

1. Extending Thread Class

```
class MyThread extends Thread{
  public void run(){
    System.out.println("Thread Running");
  }
}
class Demo{
  public static void main(String args[]){
```

```
    MyThread t=new MyThread();
    t.start();
  }
}
```

Advantages

- Easy implementation.
- Direct thread control.

Limitation

- Cannot extend another class.

2. Implementing Runnable Interface

```
class MyRunnable implements Runnable{
  public void run(){
    System.out.println("Runnable Thread");
  }
}
class Demo{
  public static void main(String args[]){
    Thread t=new Thread(new MyRunnable());
    t.start();
  }
}
```

Advantages

- Supports multiple inheritance.
- Better object-oriented design.

Suitability Analysis

Method	Suitable For
Thread Class	Small standalone thread applications
Runnable Interface	Large scalable applications

Thread Life Cycle

- New
- Runnable
- Running
- Waiting/Blocked
- Terminated

21. Summarize the Complete Event Handling Mechanism in Java AWT. [CO3]

Event Handling

Event handling is the process of responding to user actions such as mouse clicks, key presses, and button selections.

Components of Event Handling

1. Event Source

Object that generates events.

Example: Button, TextField.

2. Event Object

Contains information about the event.

Examples:

- ActionEvent
- MouseEvent
- KeyEvent
- WindowEvent

3. Event Listener

Receives and processes events.

Examples:

- ActionListener
- MouseListener
- KeyListener
- WindowListener

Event Delegation Model

1. User performs an action.
2. Event Source generates an event.
3. Event Object created.
4. Event sent to registered Listener.
5. Listener executes event-handling code.

Example

Advantages

- Separation of GUI and logic.
- Efficient event processing.
- Supports reusable components.